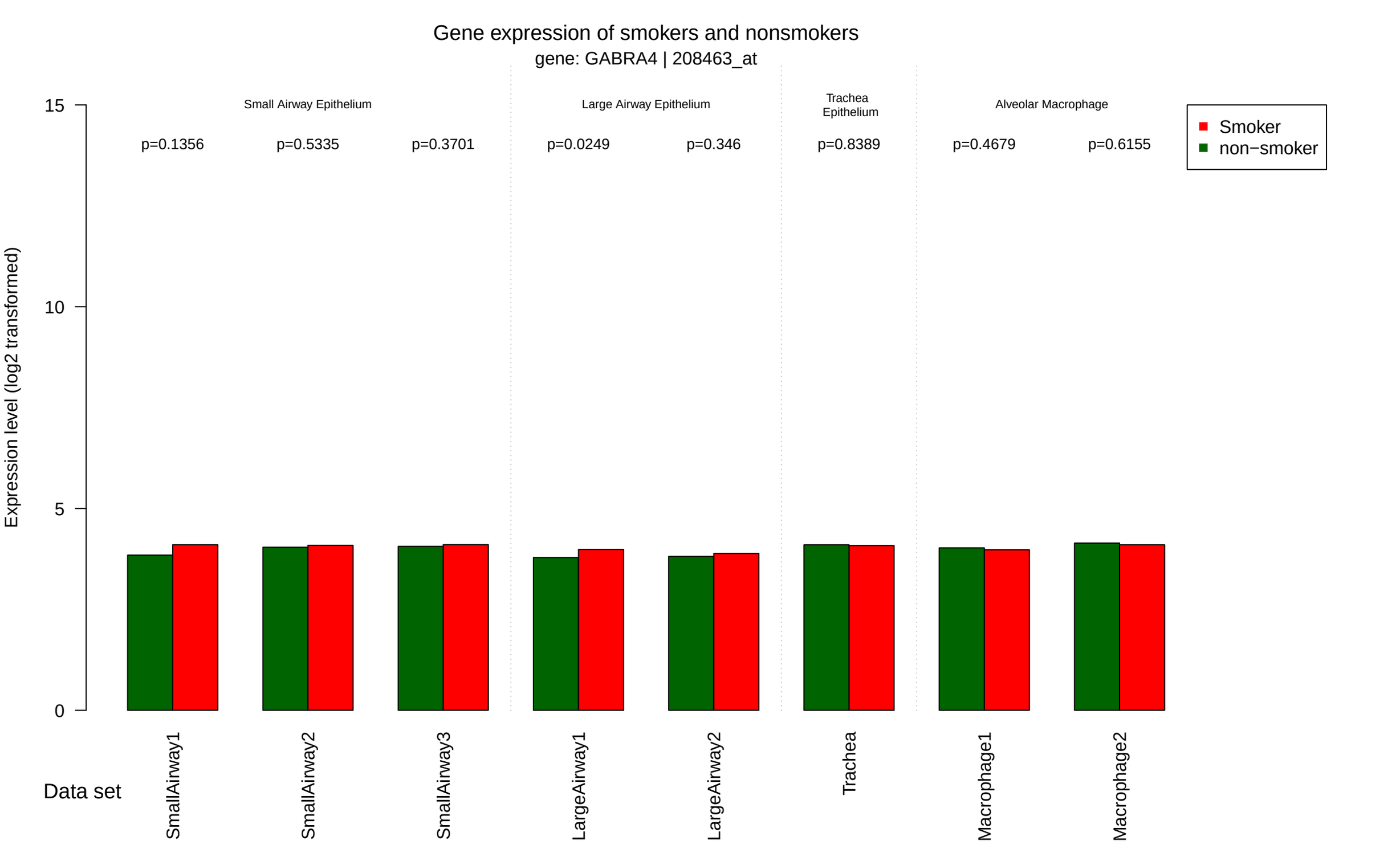
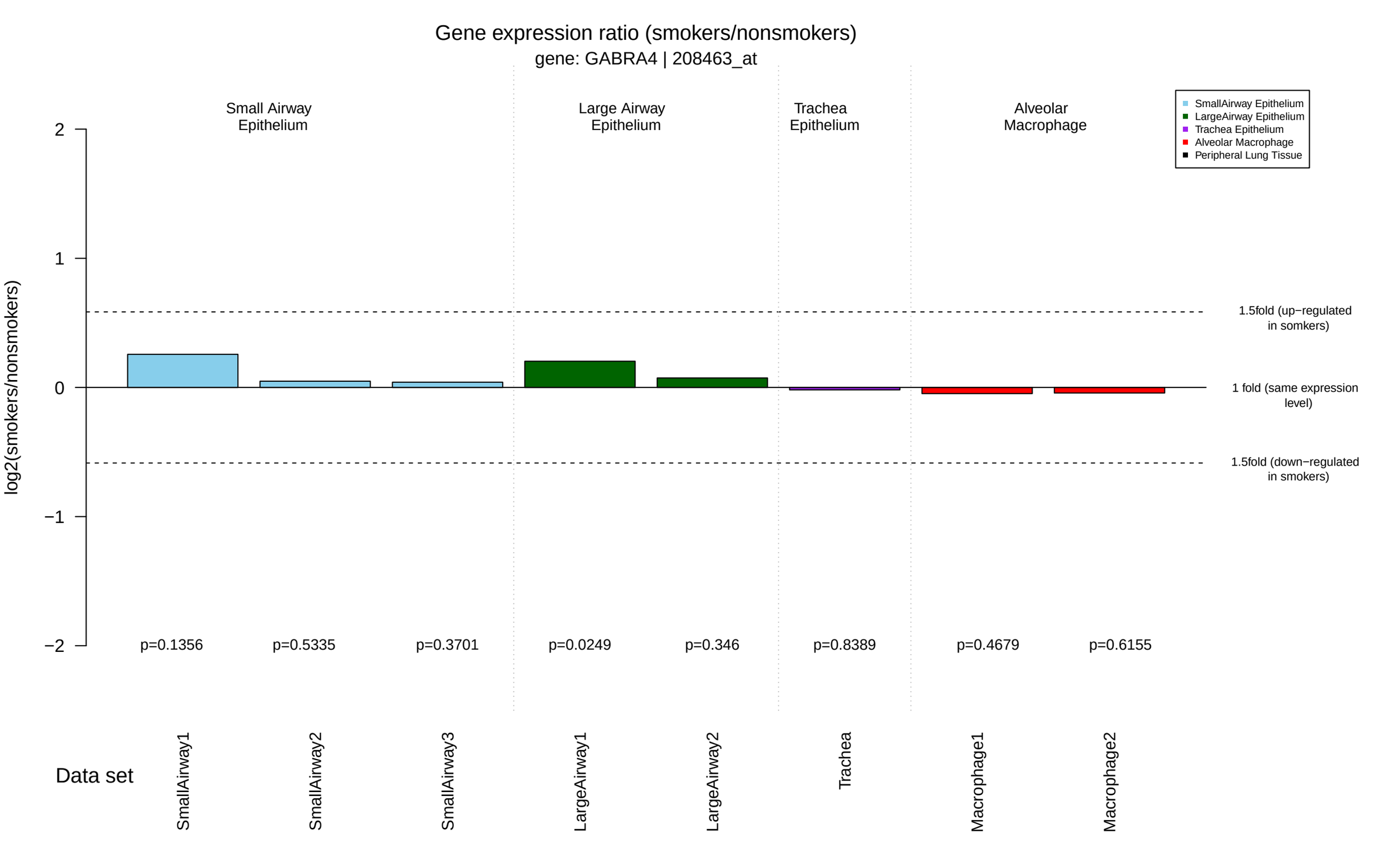
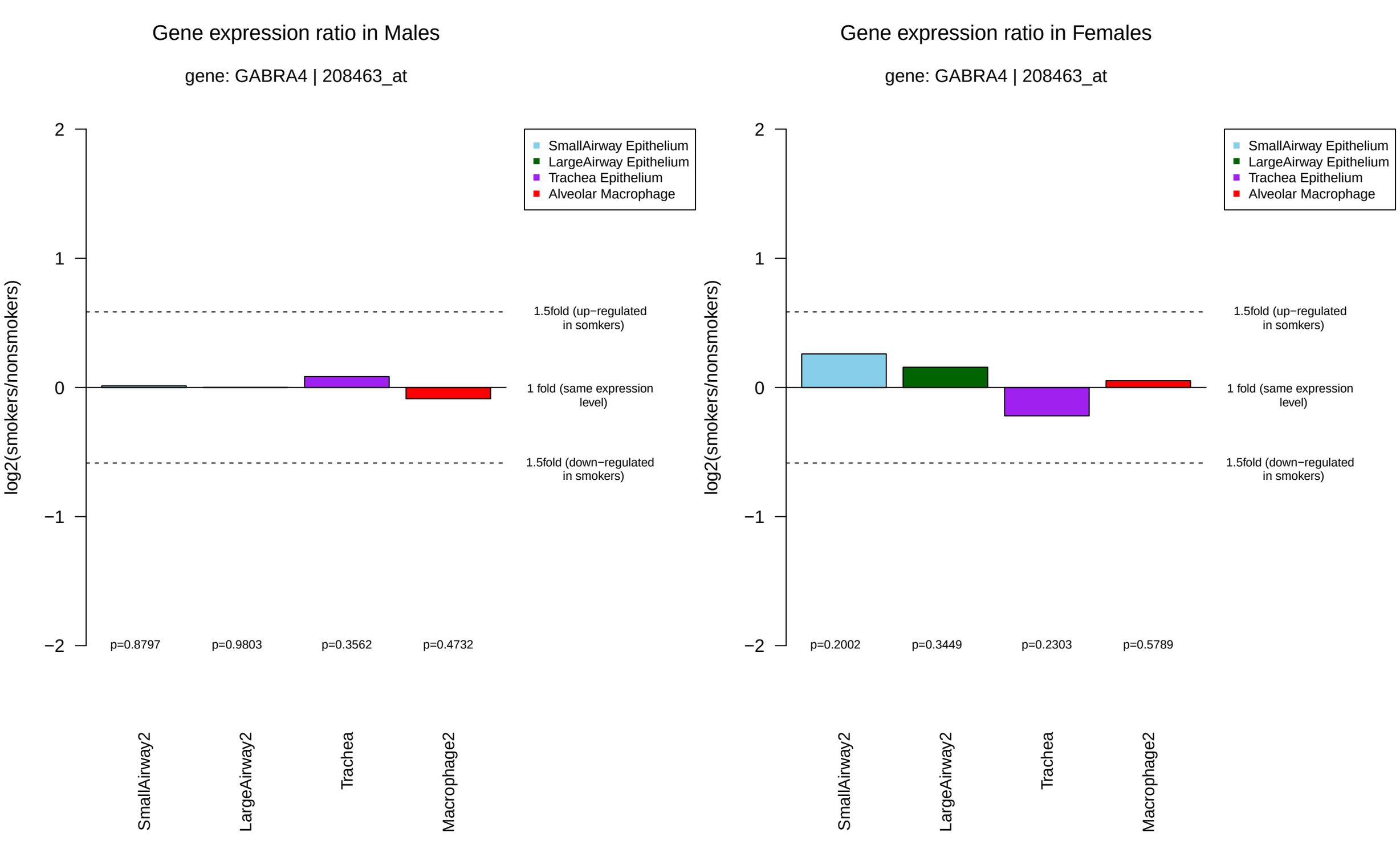
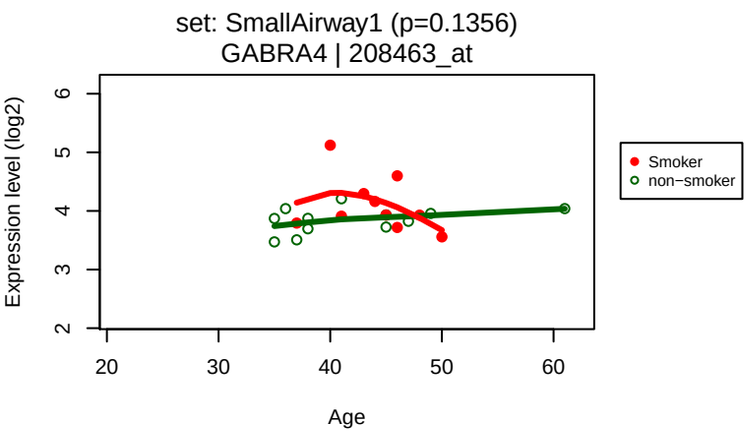
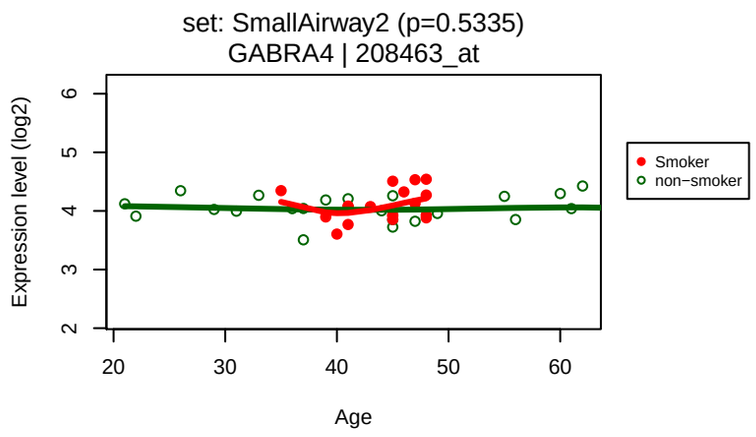
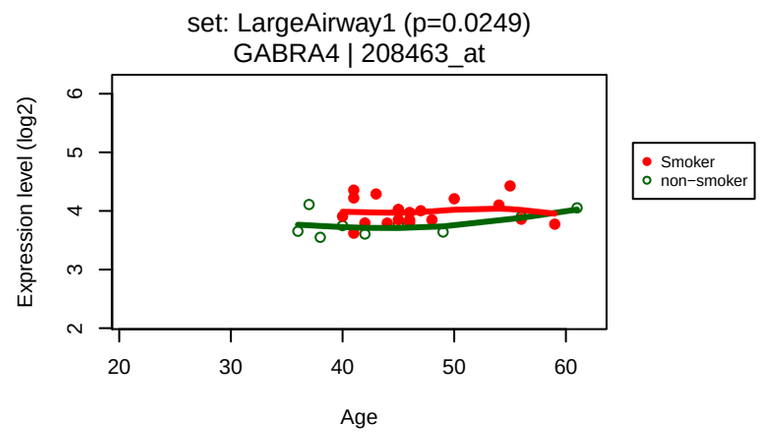

Gene expression of smokers and nonsmokers
gene: GABRA4 | 208463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1356
p=0.5335
p=0.3701
p=0.0249
p=0.346
p=0.8389
p=0.4679
p=0.6155
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

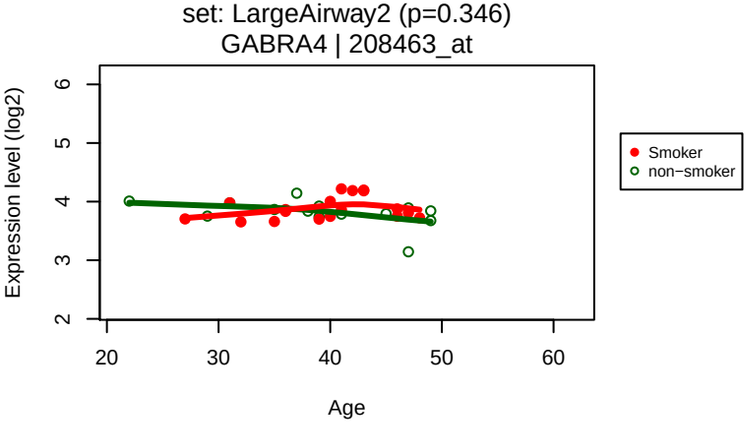

Gene expression ratio (smokers/nonsmokers)
gene: GABRA4 | 208463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1356
p=0.5335
p=0.3701
p=0.0249
p=0.346
p=0.8389
p=0.4679
p=0.6155
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

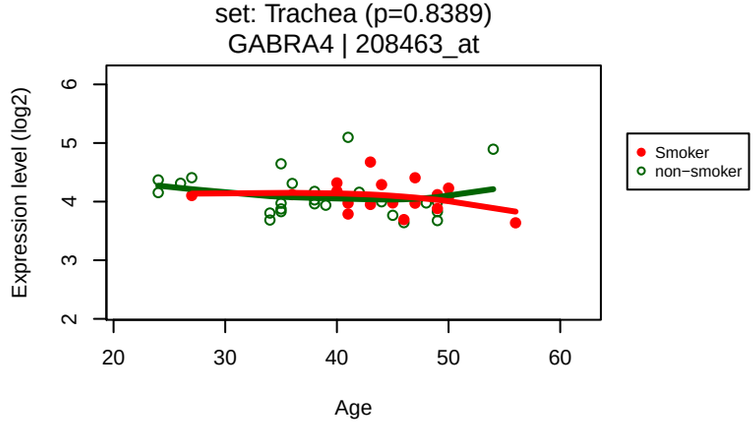

Gene expression ratio in Males
gene: GABRA4 | 208463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8797
p=0.9803
p=0.3562
p=0.4732
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GABRA4 | 208463_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2002
p=0.3449
p=0.2303
p=0.5789
SmallAirway2
LargeAirway2
Trachea
Macrophage2

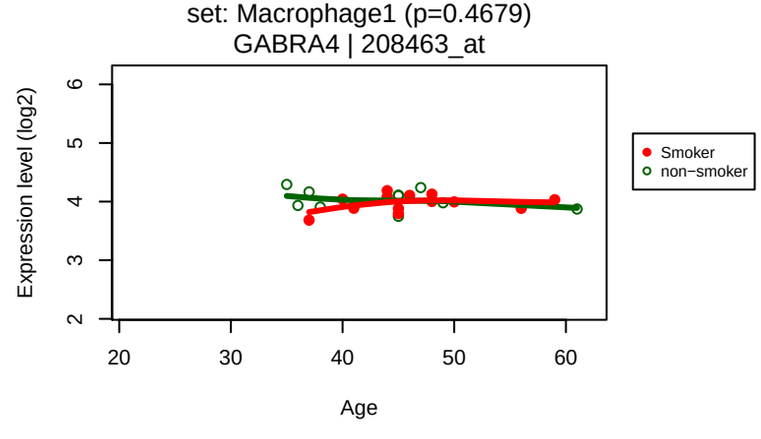

set: SmallAirway1 (p=0.1356)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker

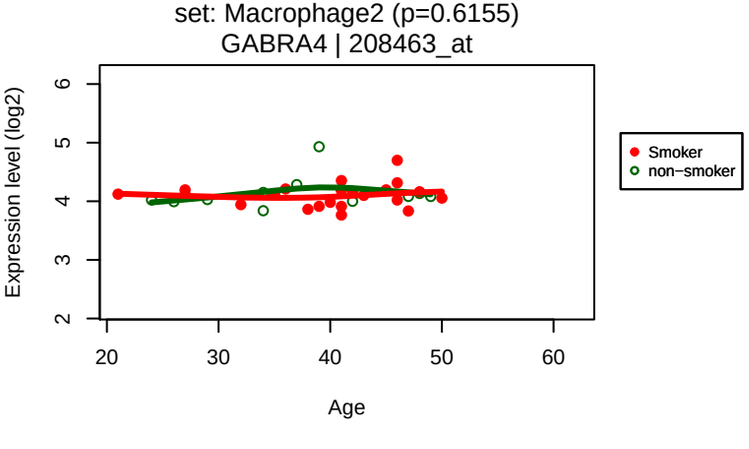

set: SmallAirway2 (p=0.5335)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker

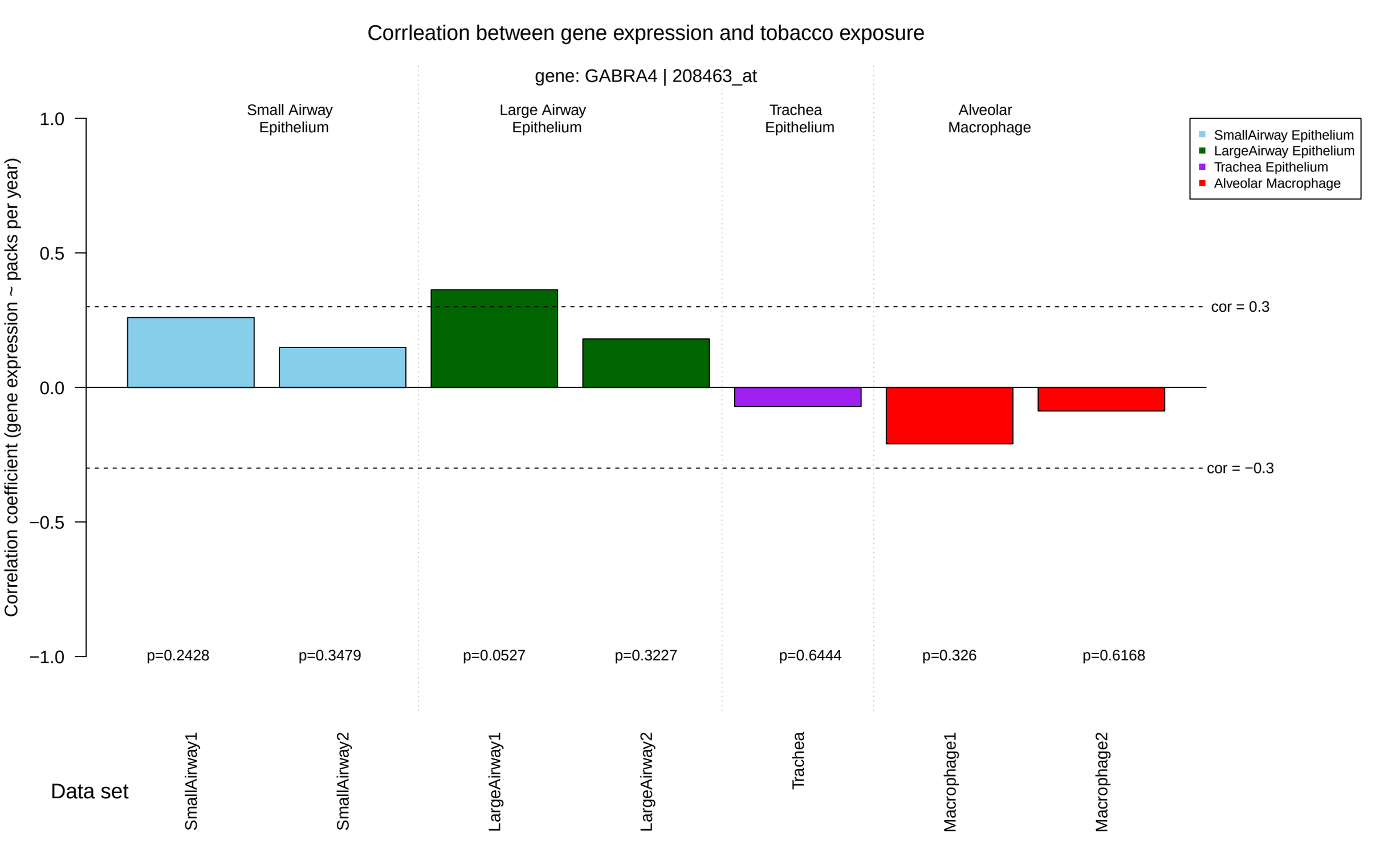

set: LargeAirway1 (p=0.0249)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.346)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8389)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4679)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6155)
GABRA4 | 208463_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GABRA4 | 208463_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2428
p=0.3479
p=0.0527
p=0.3227
p=0.6444
p=0.326
p=0.6168
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2